

Power Quality Event Source Directivity Detection Based on V-I Scatter Graph

Rodney H.G. Tan, *Member, IEEE*, and V. K. Ramachandramurthy, *Senior Member, IEEE*

Abstract – This paper presents a power quality event source directivity detection based on V-I scatter graph. The proposed method is capable to detect both voltage sag and transient power quality event source direction in term of upstream and downstream based on one bus measurement. The V-I scatter graph are plotted based on the RMS voltage and current magnitude at the same instant of time sample onto the voltage versus current graph. Based on the basic principle of power flow, the power quality event source directivity detection are determine by the sector where the plotted VI samples exceeded the voltage and current limits set on the V-I scatter graph. The proposed method is evaluated using simulation model waveforms. The evaluation shows a very promising result of power quality event source direction detection for voltage sag caused by line fault and transient caused by capacitor bank energizing.

Index Terms—Power Quality, Source Direction, Transient, Voltage Sag

I. INTRODUCTION

The rapid rise of power quality problems had significantly caught attention to both the utility and industry in the past decade. It is the concern about the quality of power being delivered to the customer. A single power quality event such as voltage sag cause by line fault may cost the affected industries up to millions of monetary loss [1] due to industry equipment failure and loss of production. Power quality events are typically categorized into voltage sag, voltage swell, transient, harmonic, voltage notch and flicker. However, non stationary power quality such as voltage sag and transient caught the highest attention due to its high number of occurrences and has immediate impact to the industry [2]. Voltage sag is typically caused by line fault at transmission or distribution level. Transient is typically caused by capacitor bank energizing for power factor correction or some other switching activities in the power system network.

Power quality monitoring system is commonly installed at the feeder bus to capture and record power quality event from time to time for power quality problem diagnosis. Most power quality monitoring system has the capability to detect and capture voltage sag or transient for recording, but does not have the capability to detect the event source direction based on one point measurement. To effectively locate the actual source location for a given power quality event, multiple PQ monitoring systems and the power system network information are require to compute the exact location. However, it is not economy or practical for

the utility company to installed power quality monitoring system at every single feeder bus or substation. It is important to identify the power quality event source direction in term of downstream or upstream from the point of monitoring to ensure mitigation and maintenance can be carried out effectively.

There are many methods being developed to detect voltage sag source direction in the last decade. There are method based on power and energy [3], method by investigating slope of the system trajectory [4], method based on phase shift jump during voltage sag [5], method based on positive sequence component [6], method based on real current [7], method based on resistance sign [8], method based on orthogonal Clarke's transformation [9], method based on wavelet multiresolution [10], method based on distance relay [11] and method based on voltage measurement only [12]. As for transient source detection, there are literatures reported on method based on disturbance energy [13] and complex wavelet based energy [14]. All the above mentioned methods have their own distinct advantage and limitation. However, the above mentioned methods were solely developed for either voltage sag or transient only, it cannot be used to detect both voltage sag and transient source direction.

This paper presents a relatively simple method based on V-I scatter graph which can be used to detect the source direction for voltage sag and transient.

II. METHODOLOGY

The proposed power quality source direction detection system is shown in Fig. 1. The RMS analysis block converts the recorded instantaneous voltage and current waveform to RMS waveform. The V-I scatter graph block plot the VI samples obtain from the RMS magnitude of the voltage and current waveform onto the voltage versus current graph. The source directivity detection block determine the power quality event source direction in term of upstream or downstream based on the conditions set in this block.

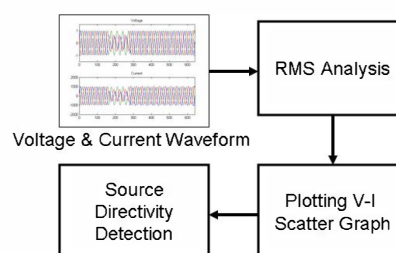


Fig. 1. Power quality source direction detection system overview

Rodney H.G. Tan is with UCSI University, Jalan Menara Gading, UCSI Height, 56000, Kuala Lumpur, Malaysia.(e-mails:rodneytan@ieee.org).

V. K. Ramachandramurthy is the head of Power Quality Research Group, Universiti Tenaga Nasional, Jalan IKRAM-UNITEN, 43000, Kajang, Selangor, Malaysia.(e-mails:vigna@uniten.edu.my).

A. RMS Analysis

The RMS analysis block converts the recorded voltage and current instantaneous waveform into RMS waveform using (1) and (2) respectively, where N is the size of the sliding window which consists the total number of samples per cycle, $v(t)$ is the voltage waveform, $i(t)$ is the current waveform, and n is the time sample index.

$$V_{rms}(n) = \sqrt{\frac{1}{N} \sum_{t=n}^{n+N-1} v^2(t)} \quad (1)$$

$$I_{rms}(n) = \sqrt{\frac{1}{N} \sum_{t=n}^{n+N-1} i^2(t)} \quad (2)$$

The RMS analysis is based on 1 period window length to prevent oscillation caused by even harmonic. Equation (1) and (2) is then normalized to per unit.

B. V-I Scatter Graph

The V-I scatter graph are plot using (3) where the position of a $VI(n)$ is determine by the RMS magnitude of the voltage $V_{rms}(n)$ and current $I_{rms}(n)$ at the same instant time sample of n plotted on the voltage versus current axis.

$$VI(n) = \{V_{rms}(n), I_{rms}(n)\} \quad (3)$$

The $VI(n)$ samples with magnitude of 1pu for both voltage and current during normal condition will be concentrated at the center position of 1pu at the voltage and current axis. The $VI(n)$ samples with abnormal magnitude caused by voltage sag or transient will be scattered outside the center position of 1pu. Since the entire RMS voltage and current waveform will be plotted on the V-I graph. The total number of $VI(n)$ sample plotted on the V-I graph is equal to the total number of time sample for the given RMS voltage and current waveform.

C. Source Directivity Detection

The power quality source direction and its relationship with the fault and monitor location is shown in Fig. 2.

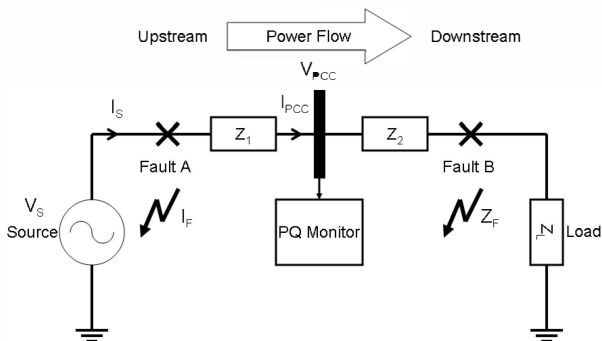


Fig. 2. Power quality event source direction and their relationship with fault and monitor location at PCC

The proposed V-I scatter graph method is based on the principle of power flow to determine the power quality event source direction. Equations (4) and (5) show the voltage and current at PCC for downstream fault, where V_{PCC} decreases as fault impedance Z_F decreases and I_{PCC}

increases as fault impedance Z_F decreases. It is also observed that for downstream fault the current at PCC I_{PCC} is equal to the source current I_S .

$$V_{PCC} = I_{PCC}(Z_2 + Z_F) \quad (4)$$

$$I_{PCC} = \frac{V_S}{Z_1 + Z_2 + Z_F} \quad (5)$$

Equations (6) and (7) show the voltage and current at PCC for upstream fault. V_{PCC} decreases as fault current I_F increases and I_{PCC} decreases as fault current I_F increases.

$$V_{PCC} = \frac{I_S - I_F}{Z_2 + Z_L} \quad (6)$$

$$I_{PCC} = I_S - I_F \quad (7)$$

The source directivity detection can be determined by the number of $VI(n)$ samples that exceeded the limit set on the V-I scatter graph. For voltage sag source direction detection, the total number of $VI(n)$ samples in the lower and upper sag current limit must first be determined in (8) and (9) respectively. For lower sag current limit in (8) where for all sample n , if the $V_{rms}(n)$ is less than 0.9pu and $I_{rms}(n)$ is less than or equal to 0.95pu, then L is increment by 1. Similarly, for upper sag current limit in (9) where for all sample n , if the $V_{rms}(n)$ is less than 0.9pu and $I_{rms}(n)$ is greater than or equal to 1.05pu, then U is increment by 1.

$$L = \{L + 1 : \forall n \text{ if } V_{rms}(n) \leq 0.9 \cap I_{rms}(n) \leq 0.95\} \quad (8)$$

$$U = \{U + 1 : \forall n \text{ if } V_{rms}(n) \leq 0.9 \cap I_{rms}(n) \geq 1.05\} \quad (9)$$

After determining the total number of $VI(n)$ samples in fall in the upper and lower sag current limit, the conditions in (10) and (11) is use to determine the voltage sag source direction. In condition (10), where the lower sag current limit L is greater than the upper sag current limit U indicates that the power is being drawn away from the feeder bus during fault in front of the PQ monitor, therefore the voltage sag source direction is upstream. On the other hand in condition (11), where the upper sag current limit U is greater than the lower sag current limit L indicates that excessive power is being drawn from the source to the feeder bus during fault behind the PQ monitor, therefore the voltage sag source direction is downstream.

$$L > U \rightarrow \text{Upstream} \quad (10)$$

$$U > L \rightarrow \text{Downstream} \quad (11)$$

For transient source direction detection, similarly the total number of $VI(n)$ samples in upper transient current limit must first be determined in (12), where for all sample n , if the $V_{rms}(n)$ is greater or equal to 0.9pu and $I_{rms}(n)$ is greater or equal to 1.05pu, then U_T is increment by 1.

$$U_T = \{U_T + 1 : \forall n \text{ if } V_{rms} \geq 0.9 \cap I_{rms}(n) \geq 1.05\} \quad (12)$$

After determining the total number of $VI(n)$ samples in fall in the upper transient current limit, the conditions in

(13) and (14) is use to determine the transient source direction. In condition (13), where the upper transient current limit U_T is equal to zero indicates that power is being drawn away from the feeder bus during capacitor bank energizing in front of the PQ monitor, therefore the transient source direction is upstream. On the other hand in condition (14), where the upper transient current limit U_T is greater than zero, again indicates that excessive power is being drawn from the source to the feeder bus during capacitor bank energizing behind the PQ monitor, therefore the transient source direction is downstream.

$$U_T = 0 \rightarrow \text{Upstream} \quad (13)$$

$$U_T > 0 \rightarrow \text{Downstream} \quad (14)$$

With the above limits and conditions, the V-I graph is basically divided into four sectors as summarized and shown in Fig. 3. The voltage sag limit of 0.9pu basically divided the voltage sag and transient sectors.

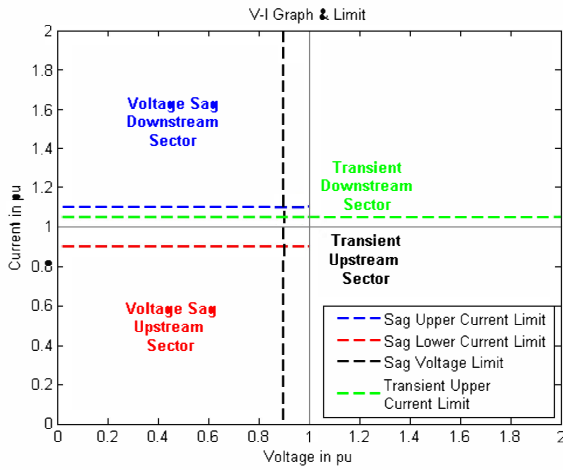


Fig. 3. The V-I graph and its respective sectors and limits for voltage sag and transient

III. RESULTS

To evaluate and demonstrate the proposed source direction detection method, a distribution line fault and capacitor bank energizing simulation models were developed to simulate voltage sag and transient using Simulink.

A. Voltage Sag Source Direction Detection

To demonstrate voltage sag source direction detection capability, a distribution line fault model used to simulate voltage sag caused by line fault was developed in Simulink as shown in Fig. 4. The distribution line fault model consists of 11kV 50Hz three phase source block feeding through 11kV/415V 1MVA delta/wye transformers to a 500kW resistive and 100kVAr inductive load. There is a fault block located behind the 11kV bus to simulate line fault. There are voltage and current scopes set with 1.6kHz sampling frequency located at 11kV and 415V buses for measurement.

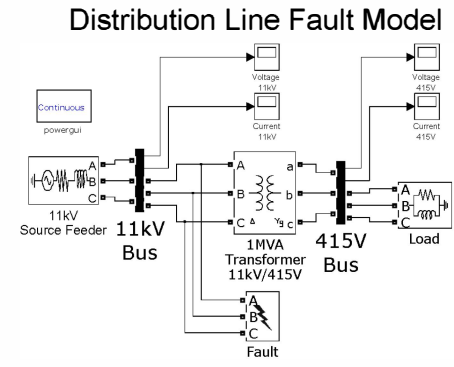


Fig. 4. Distribution line fault Simulink model to simulate line fault behind 11kV feeder bus

Fig. 5 shows the voltage and current measured and recorded at 11kV and 415V bus due to single line to ground fault at Phase A.

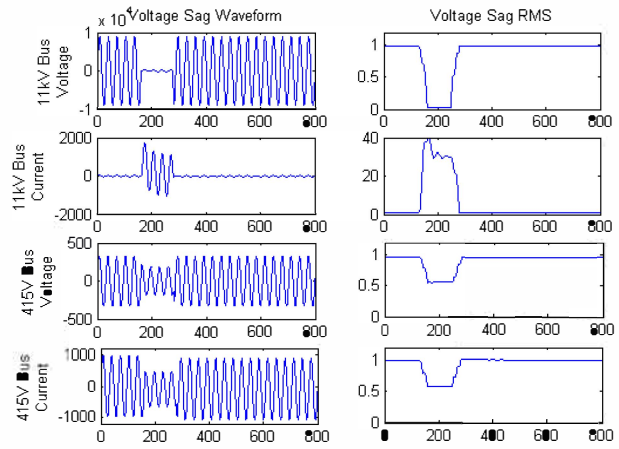


Fig. 5. Voltage sag and current waveform and RMS waveform at 11kV and 415V bus

Fig. 6 shows the V-I scatter graph for the RMS current and voltage waveform from 11kV bus. The V-I scatter graph yield a total of 141 $VI(n)$ samples that exceeded the sag current upper limit U during voltage sag indicating excessive current is being drawn from the 11kV source feeder due to downstream fault. Therefore, the voltage sag is originated from downstream.

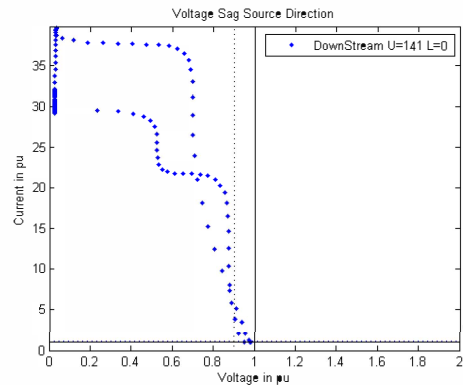


Fig. 6. V-I scatter graph for voltage sag measured at 11kV bus

Fig. 7 shows the V-I scatter graph for the RMS current and voltage waveform from 415V bus. The V-I scatter graph yield a total of 143 $VI(n)$ samples that exceeded the sag current lower limit L during voltage sag indicating current

has been drawn away from the 415V bus due to upstream fault. Therefore, the voltage sag is originated from upstream.

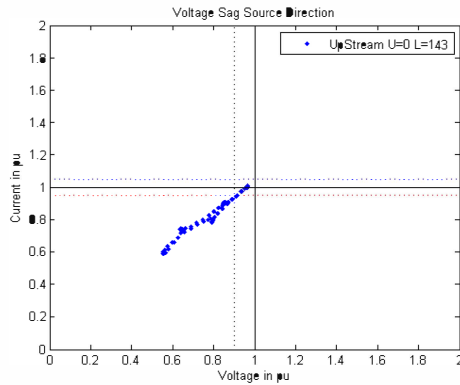


Fig. 7. V-I scatter graph for voltage sag measured at 415V bus

B. Transient Source Direction Detection

To demonstrate transient source direction detection capability, a capacitor bank energizing model used to simulate transient caused by capacitor bank energizing was developed in Simulink as shown in Fig. 8. The model consists of 11kV 50Hz three phase source block feeding through 11kV/415V delta/wye transformers to a 50kW resistive and 100kVAr inductive load. There are voltage and current scopes set with 25.6kHz sampling frequency located at 11kV and 415V buses for measurement. The 11kV feeder line consist of a capacitor bank connect through a contactor block. This capacitor bank energizing model is capable of simulating voltage transient caused by energizing of capacitor bank.

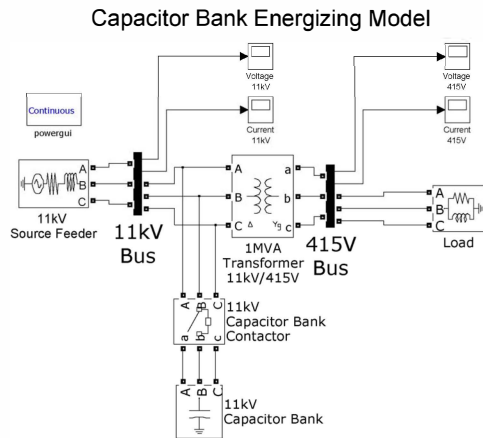


Fig. 8. Capacitor bank energizing Simulink model to simulate transient behind 11kV feeder bus

Fig. 9 shows the transient voltage and current waveform and RMS measured and recorded at 11kV and 415V bus due capacitor bank energizing behind 11kV bus.

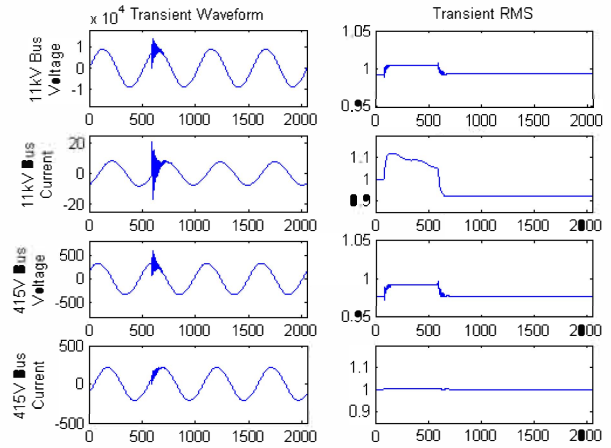


Fig. 9. Transient voltage and current waveform and RMS waveform at 11kV and 415V bus

Fig. 10 shows the V-I scatter graph for the RMS current and voltage waveform from 11kV bus. The V-I scatter graph yield a total of 504 $VI(n)$ samples that exceeded the transient current upper limit during the transient indicating excessive current is being drawn from the 11kV source feeder due to downstream capacitor bank energizing. Therefore, the transient is originated from downstream.

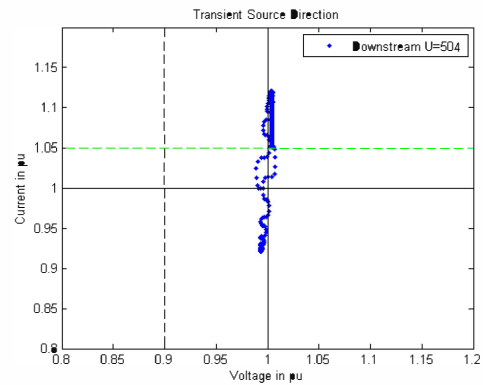


Fig. 10. V-I scatter graph for transient measured at 11kV bus

Fig. 11 shows the V-I scatter graph for the RMS current and voltage waveform from 11kV bus. The V-I scatter graph does not yield any $VI(n)$ sample that exceeded the transient current upper limit during the transient indicating current is drawn away from the 415V bus due to upstream capacitor bank energizing. Therefore, the transient is originated from upstream.

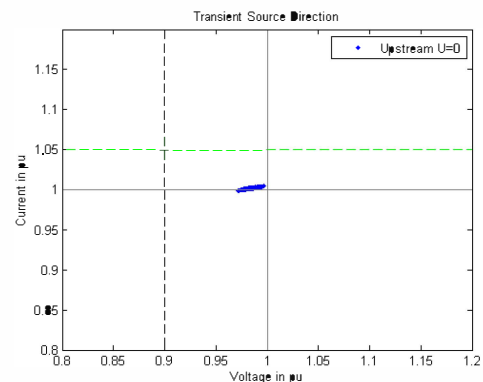


Fig. 11. V-I scatter graph for transient measured at 415V bus

IV. DISCUSSION

The performance and capability of the proposed source direction detection method was clearly demonstrated in Section III. However, it still has some limitation and rooms for improvement. The power quality event of either voltage sag or transient must first be identified before proceed to the proposed method to detect its source direction to avoid confusion and ensure accurate source direction detection. Transient source direction detection is limited to only transient caused by capacitor bank energizing. The proposed method is limited to source direction in term of upstream and downstream only. The source exact location cannot be determined. However, further research can be carried out to study the V-I scatter characteristic which may leads to the development of method to estimate the exact source location of the power quality event.

V. CONCLUSION

This paper presents a simple method to detect the source direction for voltage sag and transient based on V-I scatter graph. Voltage sag and transient simulation models were developed to simulated voltage sag and transient to evaluate the proposed V-I scatter graph method. The evaluation shows promising results in term of source direction detection for voltage sag caused by line fault and transient caused by capacitor bank energizing. By accurately detects the power quality event source direction, mitigation and maintenance can be carried out effectively. The proposed method contributes to the advancement of power quality diagnosis software use in monitoring system.

VI. REFERENCES

- [1] Mohd Izhwan Muhamad, Norman Mariun, Mohd Amran Mohd Radzi, "The Effects of Power Quality To The Industries", *IEEE 5th Student Conference on Research and Development, SCOREd.*, 2007.
- [2] Christopher J. Melhorn, Timothy D. Davis, George E. Beam, "Voltage Sags: Their Impact on the Utility and Industrial Customers", *IEEE Transactions on Industry Applications*, vol. 34, no.3, pp. 549-558, May/June 1998.
- [3] Antony C. Parsons, W. Mack Grady, Edward J. Powers, John C. Soward, "A Direction Finder for Power Quality Disturbances Based Upon Disturbance Power and Energy", *IEEE Transactions on Power Delivery*, vol. 15, no. 3, pp. 1081-1086, July 2000.
- [4] C. Li, T. Tayjasanant, W. Xu, X. Liu, "Method for Voltage Sag Source Detection by Investigating Slope of the System Trajectory", *IEE Proceeding on Generation, Transmission and Distribution*, vol. 150, no. 3, pp. 367-372, May 2003.
- [5] Juan Gomez, Medhat Morcos, Daniel Tourn, Marcos Felici, "A Noval Methodology to Locate Originating Points of Voltage Sags in Electric Power Systems", 18th International Conference on Electricity Distribution (CIRED)", June 2005.
- [6] Bostjan Polajzer, Gorazd Stumberger, Drago Dolinar, "Evaluation of Different Methods for Voltage Sag Source Detection Based on Positive-Sequence Components", International Conference on Renewable Energies and Power Quality (ICREPQ'09), April 2009.
- [7] N. Hamzah, A. Mohamed, A. Hussain, "A New Approach to Locate the Voltage Sag Source using Real Current Component", *Electric Power Systems Research*, vol. 72, no. 2, pp. 113-123, December 2004.
- [8] T. Taujasanant, V. Li, W. Xu, "A Resistance Sign-Based Method for Voltage Sag Source Detection", *IEEE Transactions on Power Delivery*, vol. 20, no. 4, pp. 2544-2552, October 2005.
- [9] B. Polajzer, G. Stumberger, S. Seme, D. Dolinar, "Detection of Voltage Sag Sources Based on Instantaneous Voltage and Current Vectors and Othogonal Clarke's Transformation", *IET Proceeding on Generation, Transmission and Distribution*, vol. 2, no. 2, pp. 219-226, March 2008.
- [10] Zhang Xue-Meng, Xu Yong-Hai, "Analysis of Voltage Sag Source Location Based on Wavelet-Multiresolution Method", Asia Pacific Power and Energy Engineering Conference (APPEEC), March 2010.

- [11] Ashok Kumar Pradhan, Aurobinda Routray, "Applying Distance Relay for Voltage Sag Source Detection", *IEEE Transactions on Power Delivery*, vol. 20, no. 1, pp. 529-531, January 2005.
- [12] Roberto Chouhy Leborgne, Daniel Karlsson, "Voltage Sag Source Location Based on Voltage Measurements Only", *Electrical Power Quality and Utilisation Journal*, vol. 14, no. 1, pp. 25-30, 2008.
- [13] Antony C. Parsons, W. Mack Grady, Edward J. Powers, John C. Soward, "Rules for Locating the Sources of Capacitor Switching Disturbances", IEEE Power Engineering Society Summer Meeting, vol. 2, pp. 749-799, July 1999.
- [14] N. Hamzah, A. Mohamed, A. Hussain, "Transient Source Location using Complex Wavelet-Based Energy", European Power and Energy Systems, 2006.

VII. BIOGRAPHIES

Rodney H.G. Tan was born in Malaysia, in 1972. He received the BSc degree in Electronic and Electrical Engineering from Robert Gordon University, UK in 1994, MSc degree in Microelectronic and Information System from Liverpool John Moores University, UK in 1996. He is currently serving UCSI University as senior lecturer. His research interest includes Power Quality and Embedded System Design.

V. K. Ramachandaramurthy completed his PhD at UMIST, UK in 2001. He presently heads the Power Quality Research Group in University Tenaga Nasional, Malaysia. He also leads several research and consultancy projects and has authored numerous journals and refereed conference papers in the areas of power system, power quality, embedded generation and renewable energy.